

AQA A-Level Further Mathematics (7367)

Question Paper

Continuous Random Variables

SD2 –Find probability for interval

(A-Level)

Name: _____

Class: _____

Date: _____

Time:

688 min.

Marks:

561 marks

Comments:

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Q1.

Participants in a school jumping competition gain a total score for each jump based on the length, L metres, jumped beyond a fixed point and a mark, S , for style.

L may be regarded as a continuous random variable with probability density function

$$f(l) = \begin{cases} wl & 0 \leq l \leq 15 \\ 0 & \text{otherwise} \end{cases}$$

where w is a constant.

S may be regarded as a discrete random variable with probability function

$$P(S = s) = \begin{cases} \frac{1}{15}s & s = 1, 2, 3, 4, 5 \\ 0 & \text{otherwise} \end{cases}$$

Assume that L and S are independent.

The total score for a participant in this competition, T , is given by $T = L^2 + \frac{1}{2}S$

Show that the expected total score for a participant is $114\frac{1}{3}$

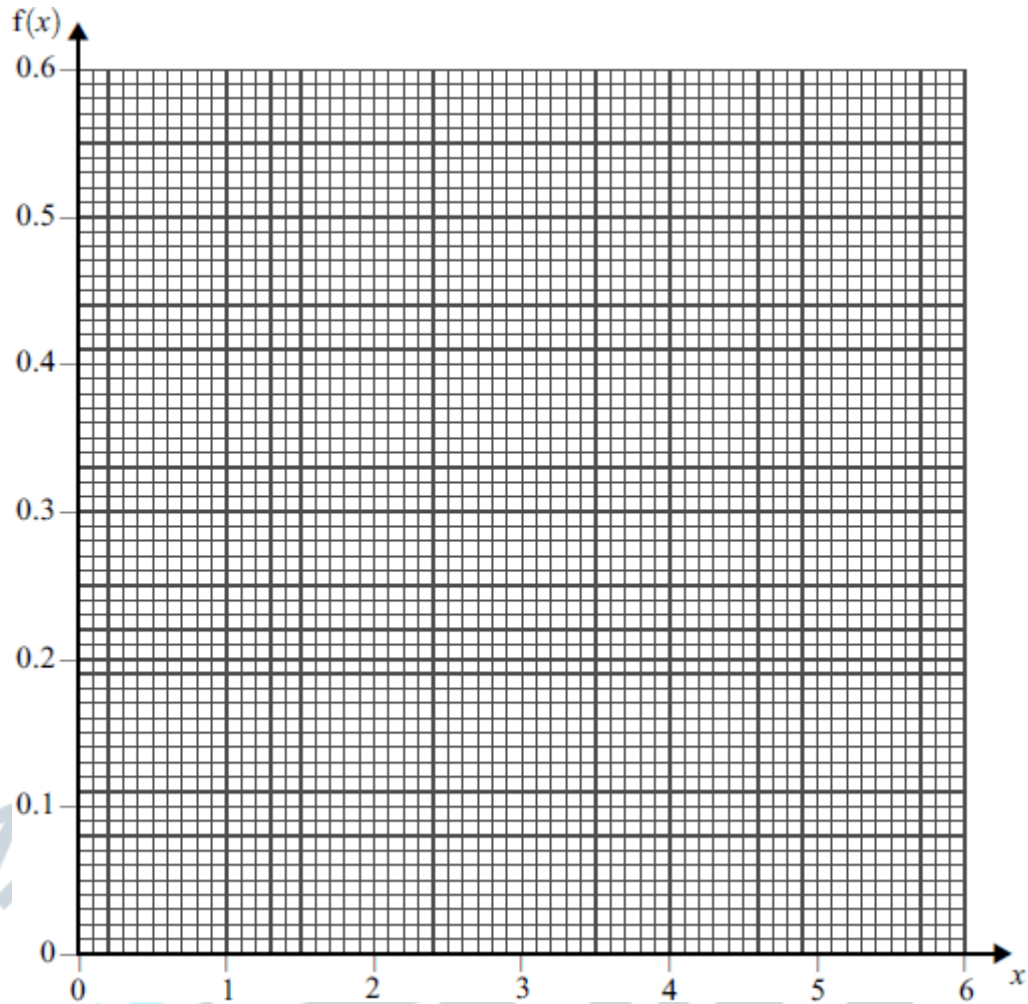
(Total 5 marks)

Q2.

A continuous random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{1}{6}(4-x) & 1 \leq x \leq 3 \\ \frac{1}{6} & 3 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

(a) Draw the graph of f on the grid below.



- (b) Prove that the mean of X is $2\frac{5}{9}$. (4)
- (c) Calculate the **exact** value of:
- (i) $P(X > 2.5)$; (2)
- (ii) $P(1.5 < X < 4.5)$; (3)
- (iii) $P(X > 2.5 \text{ and } 1.5 < X < 4.5)$; (2)
- (iv) $P(X > 2.5 \mid 1.5 < X < 4.5)$. (2)
- (Total 15 marks)

Q3.

The continuous random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{3}{8}x^2 & 0 \leq x \leq \frac{1}{2} \\ \frac{3}{32} & \frac{1}{2} \leq x \leq 11 \\ 0 & \text{otherwise} \end{cases}$$

(a) Sketch the graph of f .

(3)

(b) Show that:

$$(i) \quad P(X \geq 8\frac{1}{3}) = \frac{1}{4}$$

$$(ii) \quad P(X \geq 3) = \frac{3}{4}$$

(3)

(c) Hence write down the **exact** value of:

(i) the interquartile range of X ;

(ii) the median, m , of X .

(3)

(d) Find the **exact** value of $P(X < m \mid X \geq 3)$.

(3)

(Total 12 marks)

Q4.

The continuous random variable X has the probability density function defined by

$$f(x) = \begin{cases} \frac{3}{8}(x^2 + 1) & 0 \leq x \leq 1 \\ \frac{1}{4}(5 - 2x) & 1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

(a) The cumulative distribution function of X is denoted by $F(x)$.

Show that, for $0 \leq x \leq 1$,

$$F(x) = \frac{1}{8}x(x^2 + 3)$$

(3)

- (b) Hence, or otherwise, verify that the median value of X is 1.

(2)

- (c) Show that the upper quartile, q , satisfies the equation $q^2 - 5q + 5 = 0$ and hence that $q = \frac{1}{2}(5 - \sqrt{5})$.

(5)

- (d) Calculate the **exact** value of $P(q < X < 1.5)$.

(4)

(Total 14 marks)

Q5.

The error, in minutes, made by Paul in estimating the time that he takes to complete a college assignment may be modelled by the random variable T with probability density function

$$f(t) = \begin{cases} \frac{1}{30} & -5 \leq t \leq 25 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find:

(i) $E(T)$;

(1)

(ii) $\text{Var}(T)$.

(1)

- (b) Calculate the probability that Paul will make an error of magnitude at least 2 minutes when estimating the time that he takes to complete a given assignment.

(3)

(Total 5 marks)

Q6.

The continuous random variable X has probability density function given by

$$f(x) = \begin{cases} \frac{1}{2}(x^2 + 1) & 0 \leq x \leq 1 \\ (x - 2)^2 & 1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(3)

- (b) Calculate $P(X \leq 1)$. (3)
- (c) Show that $E(X^2) = \frac{4}{5}$. (5)
- (d) (i) Given that $E(X) = \frac{19}{24}$ and that $\text{Var}(X) = \frac{499}{k}$, find the numerical value of k . (3)
- (ii) Find $E(5X^2 + 24X - 3)$. (2)
- (iii) Find $\text{Var}(12X - 5)$. (2)
- (Total 18 marks)

Q7.

The random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{1}{2} & 0 \leq x \leq 1 \\ \frac{1}{8}(x-4)^2 & 1 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) State values for the median and the lower quartile of X . (2)
- (b) Show that, for $1 \leq x \leq 4$, the cumulative distribution function, $F(x)$, of X is given by
- $$F(x) = 1 + \frac{1}{54}(x-4)^3$$
- (You may assume that $\int (x-4)^2 dx = \frac{1}{3}(x-4)^3 + c$.) (4)
- (c) Determine $P(2 \leq X \leq 3)$. (2)
- (d) (i) Show that q , the upper quartile of X , satisfies the equation $(q-4)^3 = -13.5$. (3)
- (ii) Hence evaluate q to three decimal places. (1)

(Total 12 marks)

Q8.

The continuous random variable X has the cumulative distribution function

$$F(x) = \begin{cases} 0 & x < -c \\ \frac{x+c}{4c} & -c \leq x \leq 3c \\ 1 & x > 3c \end{cases}$$

where c is a positive constant.

- (a) Determine $P\left(-\frac{3c}{4} < X < \frac{3c}{4}\right)$. (2)

- (b) Show that the probability density function, $f(x)$, of X is

$$f(x) = \begin{cases} \frac{1}{4c} & -c \leq x \leq 3c \\ 0 & \text{otherwise} \end{cases}$$
 (2)

- (c) Hence, or otherwise, find expressions, in terms of c , for:

(i) $E(X)$; (1)

(ii) $\text{Var}(X)$. (1)

(Total 6 marks)

Q9.

The continuous random variable X has the probability density function given by

$$f(x) = \begin{cases} \frac{1}{16}x^3 & 0 \leq x \leq 2 \\ \frac{1}{6}(5-x) & 2 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f . (3)

- (b) Prove that the cumulative distribution function of X for $2 \leq x \leq 5$ can be written in the form

$$F(x) = 1 - \frac{1}{12}(5-x)^2$$

(4)

- (c) Hence, or otherwise, determine $P(X \geq 3 \mid X \leq 4)$.

(5)

(Total 12 marks)

Q10.

The continuous random variable X has probability density function given by

$$f(x) = \begin{cases} \frac{1}{2} & 0 \leq x \leq 1 \\ \frac{3-x}{4} & 1 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(3)

- (b) Explain why the value of η , the median of X , is 1.

(2)

- (c) Show that the value of μ , the mean of X , is $\frac{13}{12}$.

(4)

- (d) Find $P(X < 3\mu - \eta)$.

(3)

(Total 12 marks)

Q11.

The waiting time, X minutes, for fans to gain entrance to see an event may be modelled by a continuous random variable having the distribution function defined by

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{2}x & 0 \leq x \leq 1 \\ \frac{1}{54}(x^3 - 12x^2 + 48x - 10) & 1 \leq x \leq 4 \\ 1 & x > 4 \end{cases}$$

- (a) (i) Sketch the graph of F .

(4)

- (ii) Explain why the value of q_1 , the lower quartile of X , is $\frac{1}{2}$.

(2)

- (iii) Show that the upper quartile, q_3 , satisfies $1.6 < q_3 < 1.7$.

(3)

(b) The probability density function of X is defined by

$$f(x) = \begin{cases} \alpha & 0 \leq x \leq 1 \\ \beta(x-4)^2 & 1 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

(i) Show that the **exact** values of α and β are $\frac{1}{2}$ and $\frac{1}{18}$ respectively.

(5)

(ii) Hence calculate $E(X)$.

(5)

(Total 19 marks)

Q12.

The delay, in hours, of certain flights from Australia may be modelled by the continuous random variable T , with probability density function

$$f(t) = \begin{cases} \frac{2}{15}t & 0 \leq t \leq 3 \\ 1 - \frac{1}{5}t & 3 \leq t \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

(a) Sketch the graph of f .

(3)

(b) Calculate:

(i) $P(T \leq 2)$;

(2)

(ii) $P(2 < T < 4)$.

(3)

(c) Determine $E(T)$.

(4)

(Total 12 marks)

Q13.

The continuous random variable X has cumulative distribution function

$$F(x) = \begin{cases} 0 & x < -1 \\ \frac{x+1}{k+1} & -1 \leq x \leq k \\ 1 & x > k \end{cases}$$

where k is a positive constant.

- (a) Find, in terms of k , an expression for $P(X < 0)$. (2)
- (b) Determine an expression, in terms of k , for the lower quartile, q_1 . (3)
- (c) Show that the probability density function of X is defined by

$$f(x) = \begin{cases} \frac{1}{k+1} & -1 \leq x \leq k \\ 0 & \text{otherwise} \end{cases}$$
(2)

- (d) Given that $k = 11$:
- (i) sketch the graph of f ; (2)
- (ii) determine $E(X)$ and $\text{Var}(X)$; (2)
- (iii) show that $P(q_1 < X < E(X)) = 0.25$. (2)

(Total 13 marks)

Q14.

The waiting time, T minutes, before being served at a local newsagents can be modelled by a continuous random variable with probability density function

$$f(t) = \begin{cases} \frac{3}{8}t^2 & 0 \leq t < 1 \\ \frac{1}{16}(t+5) & 1 \leq t \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f . (3)
- (b) For a customer selected at random, calculate $P(T \geq 1)$. (2)

- (c) (i) Show that the cumulative distribution function for $1 \leq t \leq 3$ is given by

$$F(t) = \frac{1}{32}(t^2 + 10t - 7)$$

(5)

- (ii) Hence find the median waiting time.

(4)

(Total 14 marks)

Q15.

The continuous random variable X has the cumulative distribution function

$$F(x) = \begin{cases} 0 & x \leq -4 \\ \frac{x+4}{9} & -4 \leq x \leq 5 \\ 1 & x \geq 5 \end{cases}$$

- (a) Determine the probability density function, $f(x)$, of X .

(2)

- (b) Sketch the graph of f .

(2)

- (c) Determine $P(X > 2)$.

(2)

- (d) Evaluate the mean and variance of X .

(2)

(Total 8 marks)

Q16.

Students are each asked to measure the distance between two points to the nearest tenth of a metre.

- (a) Given that the rounding error, X metres, in these measurements has a rectangular distribution, explain why its probability density function is

$$f(x) = \begin{cases} 10 & -0.05 < x \leq 0.05 \\ 0 & \text{otherwise} \end{cases}$$

(3)

- (b) Calculate $P(-0.01 < X < 0.02)$.

(2)

- (c) Find the mean and the standard deviation of X .

(2)

(Total 7 marks)

Q17.

- (a) A random variable X has probability density function defined by

$$f(x) = \begin{cases} k & a < x < b \\ 0 & \text{otherwise} \end{cases}$$

(i) Show that $k = \frac{1}{b-a}$.

(1)

(ii) Prove, using integration, that $E(X) = \frac{1}{2}(a+b)$.

(4)

- (b) The error, X grams, made when a shopkeeper weighs out loose sweets can be modelled by a rectangular distribution with the following probability density function:

$$f(x) = \begin{cases} k & -2 < x < 4 \\ 0 & \text{otherwise} \end{cases}$$

- (i) Write down the value of the mean, μ , of X .

(1)

- (ii) Evaluate the standard deviation, σ , of X .

(2)

- (iii) Hence find $P\left(X < \frac{2-\mu}{\sigma}\right)$.

(3)

(Total 11 marks)

Q18.

The error, R centilitres, made in measuring the contents of bottles of wine can be modelled by the rectangular distribution

$$f(r) = \begin{cases} \alpha & -0.4 < r < 0.4 \\ 0 & \text{otherwise} \end{cases}$$

where α is a positive constant.

- (a) Find the value of α .

(1)

- (b) Calculate the mean and the standard deviation of R .

(3)

- (c) For a bottle of wine selected at random, calculate the probability that the **magnitude** of the error made in measuring its contents is less than 0.3 cl.

(2)

Q19.

The continuous random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{1}{5}(2x + 1) & 0 \leq x \leq 1 \\ \frac{1}{15}(4 - x)^2 & 1 < x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(2)

- (b) (i) Show that the cumulative distribution function, $F(x)$, for $0 \leq x \leq 1$ is

$$F(x) = \frac{1}{5}x(x + 1)$$

(3)

- (ii) Hence find $P(X \geq 0.5)$.

(2)

- (iii) Verify that the lower quartile, q_1 , satisfies

$$0.5 < q_1 < 0.75$$

(3)

(Total 10 marks)

Q20.

- (a) The continuous random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{1}{b} & 0 \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

- (i) Write down $E(X)$.

(1)

- (ii) Prove, using integration, that

$$\text{Var}(X) = \frac{1}{12}b^2$$

(5)

- (b) At an athletics meeting, the error, in seconds, made in recording the time taken to complete the 10 000 metres race may be modelled by the random variable T , having the probability density function

$$f(t) = \begin{cases} 5 & -0.1 \leq t \leq 0.1 \\ 0 & \text{otherwise} \end{cases}$$

Calculate $P(|T| > 0.02)$.

(3)

(Total 9 marks)

Q21.

The continuous random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{1}{5}(2x+1) & 0 \leq x \leq 1 \\ \frac{1}{15}(4-x)^2 & 1 < x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

(a) Sketch the graph of f .

(2)

(b) (i) Show that the cumulative distribution function, $F(x)$, for $0 \leq x \leq 1$ is

$$F(x) = \frac{1}{5}x(x+1)$$

(3)

(ii) Hence write down the value of $P(X \leq 1)$.

(1)

(iii) Find the value of x for which $P(X \geq x) = \frac{17}{20}$.

(5)

(iv) Find the lower quartile of the distribution.

(4)

(Total 15 marks)

Q22.

The time, T hours, that the supporters of Bracken Football Club have to queue in order to obtain their Cup Final tickets has the following probability density function.

$$f(t) = \begin{cases} \frac{1}{5} & 0 \leq t < 3 \\ \frac{1}{45}t(6-t) & 3 \leq t \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

(a) Sketch the graph of f .

(3)

(b) Write down the value of $P(T = 3)$.

- (c) Find the probability that a randomly selected supporter has to queue for at least 3 hours in order to obtain tickets.

(1)

(2)

- (d) Show that the median queuing time is 2.5 hours.

(2)

- (e) Calculate $P(\text{median} < T < \text{mean})$.

(6)

(Total 14 marks)

Q23.

The error, X millimetres, made when the heights of prospective members of a new gym club are measured can be modelled by a rectangular distribution with the following probability density function.

$$f(x) = \begin{cases} k & -4 \leq x \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

- (a) State the value of k .

(1)

- (b) Write down the value of $E(X)$.

(1)

- (c) Calculate $P(X > 0)$.

(2)

- (d) The height of a randomly selected prospective member is measured. Find the probability that the **magnitude** of the error made exceeds 3.5 millimetres.

(3)

(Total 7 marks)

Q24.

A machine cuts lengths of water hose for washing machines. Each length cut is at least 2 metres. The extra length, X millimetres, above 2 metres maybe modelled by the following probability density function, where c is a positive constant.

$$f(x) = \begin{cases} c & 0 \leq x \leq 4 \\ \frac{c}{16}(20 - x) & 4 \leq x \leq 20 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(4)

- (b) By considering your sketch, or otherwise, find the exact value of c .

(3)

- (c) Hence determine the probability that a randomly selected length of hose is less than 2.01 metres.

(4)

(Total 11 marks)

Q25.

The continuous random variable X has the following cumulative distribution function.

$$f(x) = \begin{cases} 0 & x < 0 \\ \frac{x^3}{64} & 0 \leq x \leq 4 \\ 1 & x > 4 \end{cases}$$

- (a) Calculate $P(2 \leq X \leq 3)$.

(2)

- (b) Find the probability density function, $f(x)$, for $0 \leq x \leq 4$.

(2)

- (c) Evaluate:

- (i) $E(X)$;

(3)

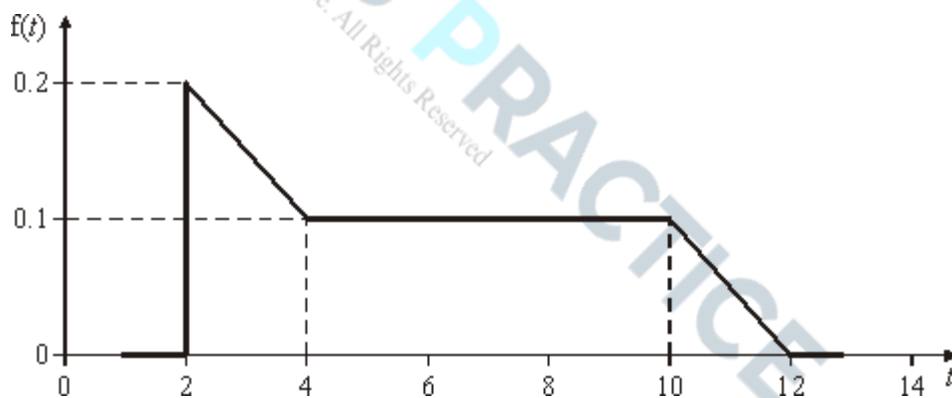
- (ii) $\text{Var}(X)$.

(3)

(Total 10 marks)

Q26.

The probability density function, $f(t)$, for the time, T minutes, taken by five-year-old children to complete a puzzle is shown by the following graph.



- (a) Determine:

- (i) $P(T < 10)$;

- (ii) $P(T > 5)$.

(3)

- (b) Hence determine the probability that a randomly selected five-year-old child completes the puzzle in less than 10 minutes, given that the child has not completed the puzzle within 5 minutes.

(4)

(Total 7 marks)

Q27.

The continuous random variable X has a rectangular distribution on the interval $(a, a + k)$, where a and k are constants and $k > 0$.

- (a) (i) Given that $\text{Var}(X) = 48$, show that $k = 24$.

(2)

- (ii) Given also that $E(X) = 6$, show that $a = -6$.

(2)

- (b) Hence calculate $P(X > 0)$.

(2)

(Total 6 marks)

Q28.

Ann and Brian usually travel to work on the same train. The train is scheduled to leave the platform at 8.00 am, but it actually leaves at time T minutes after 8.00 am. The random variable T has an exponential distribution with distribution function:

$$F(t) = \begin{cases} 1 - e^{-\frac{t}{4}} & t \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Write down the probability that the train leaves the platform at exactly 8.00 am.

(1)

- (b) One morning Ann arrives on the platform at 8.02 am and Brian arrives at 8.06 am.

- (i) Find the probability that the train has left before Ann arrives.

(2)

- (ii) Find the probability that Ann catches the train but Brian misses it.

(3)

- (c) Another morning Brian arrives on the platform at 8.01 am and is in time to catch the train. Ann arrives at 8.04 am.

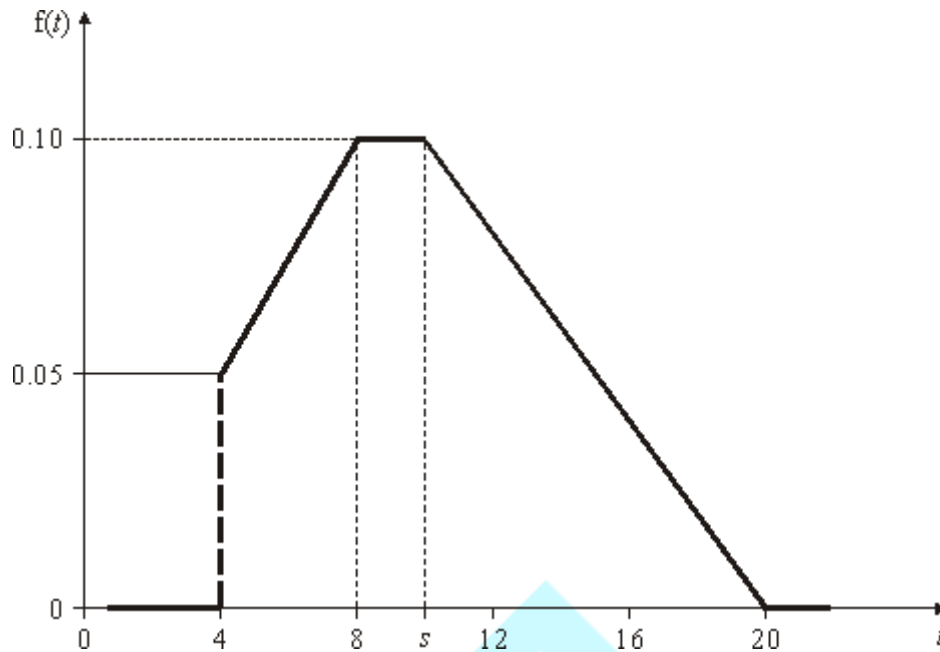
Find the probability that Ann also catches the train.

(4)

(Total 10 marks)

Q29.

At a hairdressers, the time, T minutes, required to attend to a male customer may be modelled by the probability density function, $f(t)$, as represented by the graph below.



- (a) Determine $P(T < 8)$

(2)

- (b) (i) Show why the value of s , marked on the graph above, must be 10.

(3)

- (ii) Hence determine $P(T > 15)$.

(3)

(Total 8 marks)

Q30.

The continuous random variable X has a rectangular distribution on the interval (a, b) , where $0 < a < b$.

- (a) Given that the mean, μ , is equal to 21 and that the variance, σ^2 , is equal to 27, prove that $a = 12$ and $b = 30$.

(5)

- (b) Hence determine:

- (i) $P(5 < X < 20)$;

(3)

- (ii) $P\left(X < \mu - \frac{\sigma\sqrt{3}}{2}\right)$.

(3)

(Total 11 marks)

Q31.

The time, T hours, taken by any member of a group of friends to complete a run for charity

can be modelled by the following probability density function.

$$f(t) = \begin{cases} \frac{1}{90}t^2 & 3 \leq t \leq 6 \\ 2 - \frac{4}{15}t & 6 \leq t \leq 7.5 \\ 0 & \text{otherwise} \end{cases}$$

- (a) (i) Show that the probability that a member of the group selected at random takes at least 6 hours to complete the run is 0.3.

(2)

- (ii) Evaluate the median time taken to complete the run.

(6)

- (b) Calculate the mean time taken to complete the run.

(5)

(Total 13 marks)

Q32.

A town's library offers a booking system for free internet access. Users are allowed a maximum of one hour of connection time at any one booking.

The actual connection times, X minutes, of users may be modelled by the following probability density function, where k is a constant.

$$f(x) = \begin{cases} \frac{kx}{40} & 0 \leq x \leq 40 \\ k & 40 \leq x \leq 60 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(4)

- (b) By considering your sketch, or otherwise, show that the value of k is 0.025.

(3)

- (c) Hence determine $P(X > 30)$.

(3)

(Total 10 marks)

Q33.

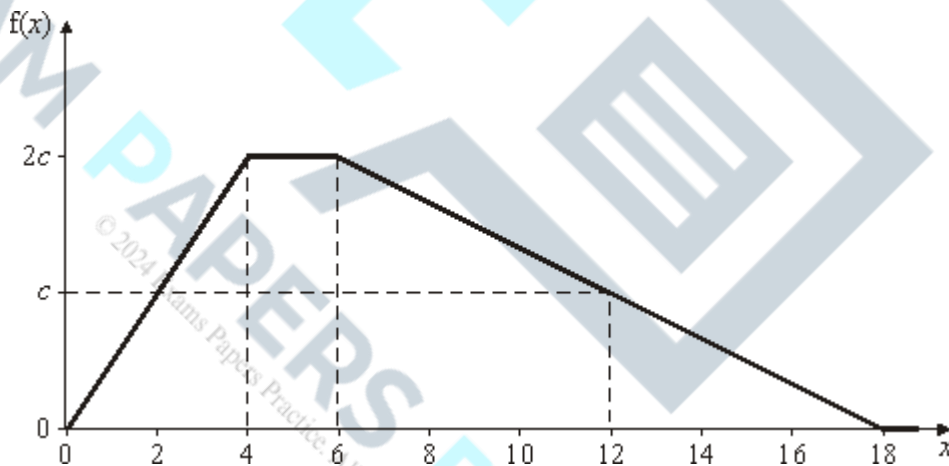
A continuous random variable T has the probability density function $f(t)$ given by:

$$f(t) = \begin{cases} \frac{1}{18}t^2 & 0 \leq t \leq 3 \\ \frac{1}{4}(5-t) & 3 \leq t \leq 5 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Sketch the graph of f . (2)
- (b) Determine the distribution function, $F(t)$, for $0 \leq t \leq 5$. (4)
- (c) Hence, or otherwise, calculate $P(T < 4)$. (2)
- (Total 8 marks)

Q34.

The time, X minutes, by which an appointment is delayed at a health centre may be modelled by a probability density function represented by the graph below.



- (a) Show that the value of the constant c , as used on the vertical axis, is 0.05. (3)
- (b) Determine the probability that an appointment is delayed:
- (i) by more than 4 minutes; (2)
 - (ii) by between 4 and 12 minutes; (3)
 - (iii) by less than 12 minutes, given that it is delayed by more than 4 minutes. (4)
- (c) Give a reason why the probability density function, as represented by the graph above, is unlikely to provide a satisfactory model for **all** delay times at this health centre.

(1)
(Total 13 marks)

Q35.

The continuous random variable X has a rectangular distribution over the interval c to $7c$, where c is a positive constant.

(a) (i) Find, in terms of c , expressions for the mean, μ , and variance, σ^2 , of X . (2)

(ii) Hence show that $E(X^2) = 19c^2$. (2)

(b) Given that $E(X^2) = 171$, determine the value of c . (1)

(c) Determine:

(i) $P\left(X > \frac{\mu}{2} + \frac{\sigma}{\sqrt{3}}\right)$, (3)

(ii) the value of d such that $P(X < d) = 0.25$. (3)
(Total 11 marks)

Q36.

The time, in minutes, that vehicles have to wait at a T-junction may be represented by a random variable X with probability density function:

$$f(x) = \begin{cases} k(2.5 - x) & 0 \leq x \leq 2.5 \\ 0 & \text{otherwise.} \end{cases}$$

(a) Sketch the graph of f . (3)

(b) By considering your sketch, or otherwise, show that the value of the constant k is 0.32. (3)

(c) Determine the probability that a vehicle has to wait at the T-junction for between 1 minute and 2 minutes. (3)
(Total 9 marks)

Q37.

The error, X grams, made by a butcher's scales, may be modelled by a continuous uniform (rectangular) distribution defined by:

$$f(x) = \begin{cases} \frac{1}{4c} & -c \leq x \leq 3c \\ 0 & \text{otherwise.} \end{cases}$$

- (a) State, in terms of c , values for the mean and variance of X .

(2)

- (b) Given that $E(X^2) = \frac{28}{3}$, show that $c = 2$.

(3)

- (c) Determine:

- (i) $P(X = 2)$;

(1)

- (ii) $P(|X| < 2.5)$.

(3)

(Total 9 marks)

Q38.

At a Post Office, the time, T minutes, that customers have to wait in order to be served has the following probability density function.

$$f(t) = \begin{cases} \frac{t^2}{18} & 0 \leq t < 3 \\ \frac{1}{4}(5-t) & 3 \leq t \leq 5 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Write down the value of $P(T = 4)$.

(1)

- (b) Show that the median waiting time is 3 minutes.

(2)

- (c) Find the probability that customers have to wait for less than 3.5 minutes to be served.

(5)

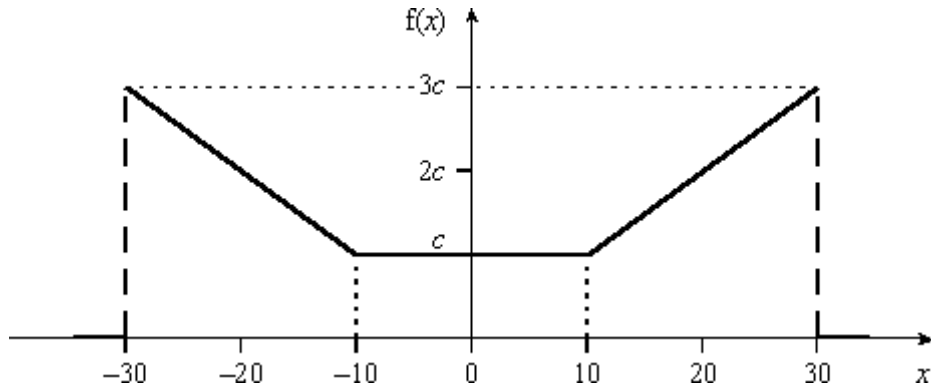
- (d) Calculate the mean time that customers have to wait in order to be served.

(5)

(Total 13 marks)

Q39.

The probability density function, $f(x)$, for the difference in millimetres from the nominal length of metal rods cut by a faulty machine, is shown by the following graph.



- (a) Show that the value of the constant c , as used on the vertical axis, is 0.01.

(3)

- (b) Determine:

(i) $P(X > 20)$;

(2)

(ii) $P(X < 10)$.

(2)

(Total 7 marks)

Q40.

All consultations with an optician are by appointment. The time, T minutes by which an appointment is delayed has the following probability density function.

$$f(t) = \begin{cases} \frac{1}{54}t^2 & 0 \leq t < 3 \\ \frac{1}{6} & 3 \leq t < 6 \\ \frac{1}{24}(10-t) & 6 \leq t < 10 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(4)

- (b) Use your graph to find:

(i) $P(3 \leq T < 6)$;

(2)

(ii) $P(T > 6)$;

(2)

(iii) the lower quartile

(4)

- (c) What proportion of appointments is delayed by less than 2 minutes?

(4)

(Total 16 marks)

Q41.

The delay, T minutes, in connecting electronically to a ticket booking service can be modelled by the distribution function:

$$P(T \leq t) = 1 - e^{-2t}, \quad t \geq 0.$$

- (a) (i) Write down the probability density function of T . (1)
- (ii) State values for the mean and standard deviation of T . (2)
- (b) Calculate the probability that the delay exceeds 3 minutes. (2)
- (c) Find the median delay, giving your answer in seconds, to the nearest second. (3)
- (Total 8 marks)**

Q42.

The queuing time, X minutes, at a post office maybe modelled by the following probability density function, where c is a constant.

$$f(x) = \begin{cases} c & 0 \leq x \leq 1 \\ \frac{c}{2}(3-x) & 1 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f . (4)
- (b) By considering your sketch, or otherwise, find the value of c . (3)
- (c) Determine $P(X < 2)$. (3)
- (Total 10 marks)**

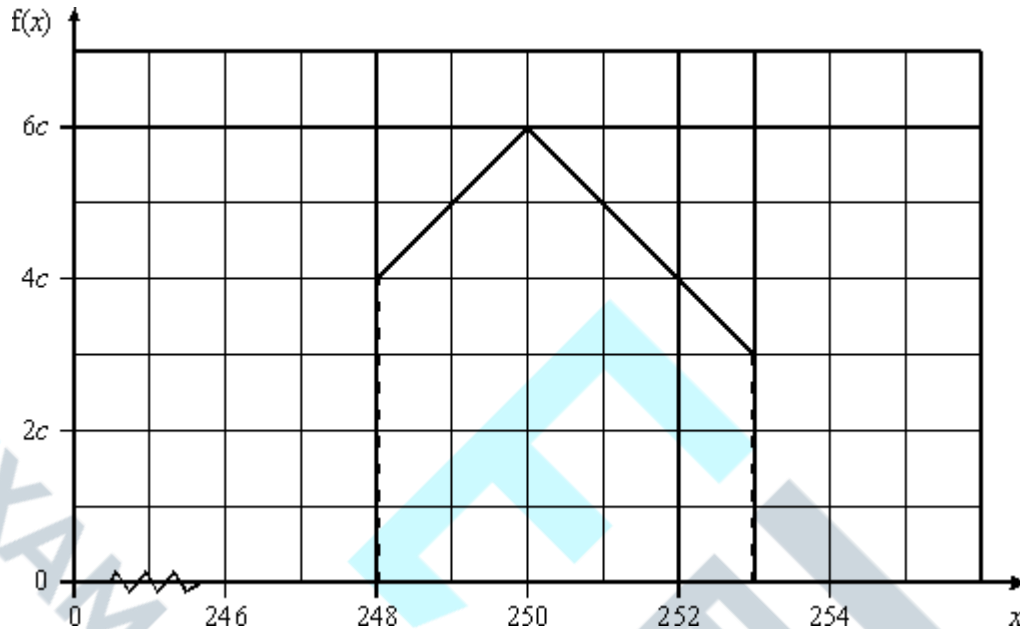
Q43.

The continuous random variable X has a rectangular distribution over the interval a to $a + k$, where a and k are positive constants.

- (a) Find, in terms of a and k , expressions for the mean and variance of X . (2)
- (b) Given that $E(X) = 21$ and $\text{Var}(X) = 3$, show that $k = 6$ and $a = 18$. (3)
- (c) Hence calculate $P(20 < X < 25)$

Q44.

The probability density function for the volume, X millilitres, of tonic water in small bottles is shown by the following graph.



- (a) Show that the value of the constant c , as used on the vertical axis, is $\frac{2}{47}$. (4)
- (b) State the probability that a bottle contains exactly 250 ml of tonic water. (1)
- (c) Determine the probability that a bottle contains between 249 ml and 252 ml of tonic water. (3)

(Total 8 marks)

Q45.

The lengths, X centimetres, of boards used in garden fencing are uniformly distributed over the interval 198 to 204.

- (a) Find values for the mean, μ , and the variance, σ^2 , of X . (2)
- (b) Write down the probability density function, $f(x)$, of X for $198 \leq x \leq 204$. (1)
- (c) Determine:
- (i) $P(X = \mu)$; (1)

(ii) $P(\bar{X} < 200 - \sigma)$.

(3)

(Total 7 marks)

Q46.

In an attempt to economise on her mobile phone costs, Tara times her calls to ensure that no call lasts longer than 6 minutes. The resultant lengths of her calls, T minutes, may be regarded as a continuous random variable with the following probability density function.

$$f(t) = \begin{cases} k(2t + 3) & 1 \leq t \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(3)

- (b) By considering your sketch, or otherwise, show that the value of the constant k is $\frac{1}{50}$.

(3)

- (c) Determine the probability that the length of a call is:

- (i) at most 3 minutes;

(3)

- (ii) between 2 and 5 minutes.

(3)

(Total 12 marks)

Q47.

The total amount of added time, S minutes, played per match by a team in a local football league is distributed with probability density function given by:

$$f(s) = \begin{cases} ks^2 & 1 \leq s \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Show that the exact value of the constant k is $\frac{3}{124}$.

(4)

- (b) Determine $P(S < 3)$.

(3)

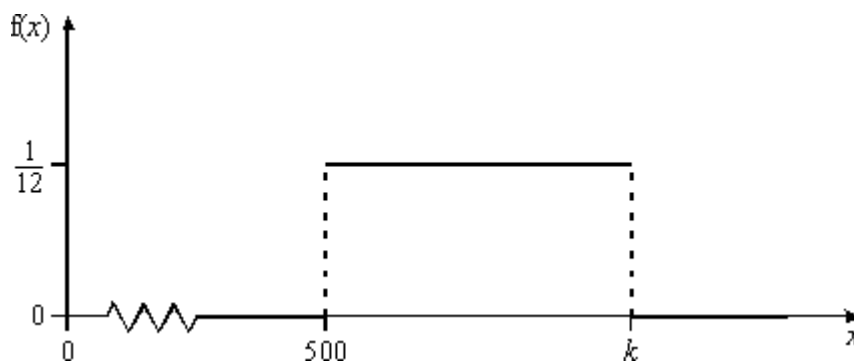
- (c) Calculate the median amount of added time played per match, giving your answer to the nearest minute.

(4)

(Total 11 marks)

Q48.

The probability density function, $f(x)$, for the weight, X grams, of marshmallow pieces in a tub is shown by the following graph.



- (a) State why the value of k on the horizontal axis is 512. (1)
- (b) Determine values for the mean, μ , and standard deviation, σ , of X . (3)
- (c) Calculate: (2)
- (i) $P(X < 509)$; (2)
- (ii) $P(X > \mu - \sigma)$. (3)
- (Total 9 marks)

Q49.

The time, H hours, to carry out a 24 000 miles service on a car may be regarded as a continuous random variable with the following probability density function.

$$f(h) = \begin{cases} c(7 - 2h) & 1 \leq h \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

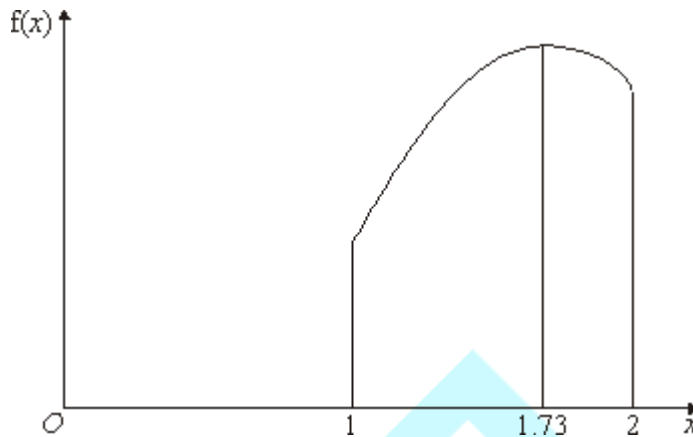
- (a) Sketch the graph of f . (3)
- (b) By considering your sketch, or otherwise, show that the value of the constant c is $\frac{1}{6}$. (3)
- (c) Calculate the probability that a service takes more than 2 hours. (2)
- (d) Given that a service is **not** completed in 2 hours, calculate the probability that it is completed in less than $2\frac{1}{2}$ hours. (5)

(Total 13 marks)

Q50.

The sketch below shows the probability density function $f(x)$ of the continuous random variable X .

($f(x) = 0$ outside the interval $1 \leq x \leq 2$.)



You are given that

$\int_1^{1.56} f(x) \, dx = 0.5$	$\int_{1.73}^2 f(x) \, dx = 0.291$
$\int_1^2 x f(x) \, dx = 1.54$	$\int_1^2 (x - 1.54)^2 f(x) \, dx = 0.0751$

(a) Write down the value of:

(i) $\int_1^2 f(x) \, dx$;

(ii) the mean of X ;

(iii) the median of X ;

(iv) the mode of X ;

(v) the range of X .

(5)

(b) Find:

(i) the probability that X lies between the median and the mode;

(2)

(ii) the standard deviation of X ;

(2)

(iii) $E(X^2)$.

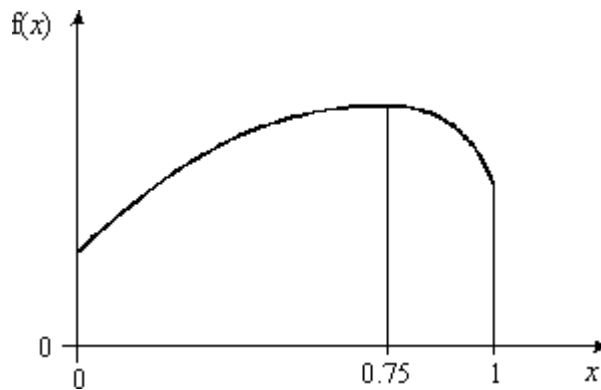
(2)

(Total 11 marks)

Q51.

The sketch below shows the probability density function $f(x)$ of the continuous random variable X .

$[f(x) = 0 \text{ outside the interval } 0 \leq x \leq 1]$



- (a) Explain why the value of the mode of X is 0.75.

(1)

- (b) You are given that:

$$E(X) = 0.542$$

$$E(X^2) = 0.369$$

$$\int_0^{0.557} f(x) dx = 0.500$$

$$\int_0^{0.542} f(x) dx = 0.483$$

$$\int_0^{0.75} f(x) dx = 0.719$$

- (i) Find the standard deviation of X .

(3)

- (ii) Find the probability that X exceeds the mode.

(2)

- (iii) Write down the value of the median of X .

(1)

- (iv) Evaluate the probability that X lies between the median and the mean.

(3)

(Total 10 marks)

Q52.

The errors, in grams, made by a butcher's scales may be modelled by a random variable, y , with probability density function

$$f(y) = \begin{cases} k & -6 < y < 14 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Show that $k = 0.05$.

(2)

- (b) Write down the mean value of Y .

- (1)
- (c) (i) Use integration to find $E(Y^2)$. (3)
- (3)
- (ii) Hence, or otherwise, find the standard deviation of Y . (3)
- (3)
- (d) (i) Find the probability that an error is negative. (1)
- (1)
- (ii) Show that the probability of an error having magnitude less than 2 is 0.2. (2)
- (2)
- (iii) Find the probability that the magnitude of an error is greater than 4. (2)
- (2)
- (Total 14 marks)**

Q53.

A supermarket sells 1 kg bags of potatoes. It is not possible to select potatoes so that each bag weighs exactly 1 kg and so the weight of each bag exceeds 1 kg by an amount, x grams. It is believed that X may be modelled by a random variable which follows a rectangular distribution with probability density function:

$$f(x) = \begin{cases} \frac{1}{c} & 0 \leq x \leq c \\ 0 & \text{otherwise} \end{cases}$$

where c is a constant.

- (a) (i) Write down the mean of this distribution in terms of c . (5)
- $\frac{c^2}{12}$
- (ii) Using integration, show that the variance is $\frac{c^2}{12}$. (5)
- (5)
- (b) Find the probability that X is greater than $\frac{c}{4}$. (2)
- (2)
- (c) The mean weight of a random sample of bags exceeds 1 kg by 14 grams. Estimate:
- (i) the standard deviation of X ; (4)
- (4)
- (ii) the weight of the heaviest bag. (1)
- (1)
- (Total 12 marks)**